

EE673 Assignment 3

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Contains the answers to the two theory questions (Q1, Q3). Please find the codes for Q2 in the zip, and instructions for running those codes in README.txt.

Question 1

(Note: The packet captures for these labs were performed on macOS. Therefore, the traceroute implementation utilizes UDP probe packets rather than the ICMP Echo Request packets natively used by Windows tracert.)

Part 1: ICMP Lab

ICMP and Ping

```
Last login: Fri Apr 10 19:38:06 on ttys004
(base) adityagautam@Adityas-MacBook-Pro cs633a2 % ping -c 10 www.ust.hk
PING www.ust.hk (143.89.209.9): 56 data bytes
64 bytes from 143.89.209.9: icmp_seq=0 ttl=32 time=390.416 ms
64 bytes from 143.89.209.9: icmp_seq=1 ttl=32 time=320.813 ms
64 bytes from 143.89.209.9: icmp_seq=2 ttl=32 time=530.731 ms
64 bytes from 143.89.209.9: icmp_seq=3 ttl=32 time=551.471 ms
64 bytes from 143.89.209.9: icmp_seq=4 ttl=32 time=331.969 ms
64 bytes from 143.89.209.9: icmp_seq=5 ttl=32 time=317.891 ms
Request timeout for icmp_seq 6
64 bytes from 143.89.209.9: icmp_seq=7 ttl=32 time=325.612 ms
64 bytes from 143.89.209.9: icmp_seq=8 ttl=32 time=318.716 ms
64 bytes from 143.89.209.9: icmp_seq=9 ttl=32 time=362.600 ms

--- www.ust.hk ping statistics ---
10 packets transmitted, 9 packets received, 10.0% packet loss
round-trip min/avg/max/stddev = 317.891/383.358/551.471/87.441 ms
(base) adityagautam@Adityas-MacBook-Pro cs633a2 %
```

1. What is the IP address of your host? What is the IP address of the destination host?

Based on the packet details, the IP address of the source host is **172.23.68.32**. The IP address of the destination host (www.ust.hk) is **143.89.209.9**.

2. Why is it that an ICMP packet does not have source and destination port numbers?

ICMP is a **Network Layer** protocol that is encapsulated directly into IP datagrams. Port numbers are a feature of **Transport Layer** protocols (like **TCP** and **UDP**) used to multiplex traffic to specific application processes. Since ICMP is used for host-to-host network control, routing, and error messaging rather than application communication, it does not utilize port numbers.

```
> Frame 4701: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface en0, id 0
> Ethernet II, Src: Apple_8b:47:44 (9c:3e:53:8b:47:44), Dst: Cisco_e8:0d:42 (64:8f:3e:e8:0d:42)
v Internet Protocol Version 4, Src: 172.23.68.32, Dst: 143.89.209.9
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 84
  Identification: 0x5a5c (23132)
> 000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: ICMP (1)
  Header Checksum: 0xcfb2 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 172.23.68.32
  Destination Address: 143.89.209.9
  [Stream index: 26]
v Internet Control Message Protocol
  Type: Echo (ping) request (8)
  Code: 0
  Checksum: 0xbe23 [correct]
  [Checksum Status: Good]
  Identifier (BE): 54564 (0xd524)
  Identifier (LE): 9429 (0x24d5)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Response frame: 4704]
v ICMP Data: 69da76ef000498e508090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f3031323334353637
  Timestamp from icmp data: Apr 11, 2026 21:59:35.301285000 IST
  [Timestamp from icmp data (relative): 190.000 microseconds]
> Data (48 bytes)
```

3. Examine one of the ping request packets sent by your host. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number and identifier fields?

The ICMP Type is **8** and the Code is **0**.

The other fields present in the ICMP header are the **Checksum**, **Identifier**, **Sequence Number**, and the **Data payload** (which contains a timestamp).

The Checksum, Identifier, and Sequence Number are each **2 bytes** (16 bits) long.

```

> Frame 4704: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface en0, id 0
> Ethernet II, Src: Cisco_e8:0d:42 (64:8f:3e:e8:0d:42), Dst: Apple_8b:47:44 (9c:3e:53:8b:47:44)
v Internet Protocol Version 4, Src: 143.89.209.9, Dst: 172.23.68.32
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0xbec4 (48836)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 32
    Protocol: ICMP (1)
    Header Checksum: 0x8b4a [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 143.89.209.9
    Destination Address: 172.23.68.32
    [Stream index: 26]
v Internet Control Message Protocol
  Type: Echo (ping) reply (0)
  Code: 0
  Checksum: 0xc623 [correct]
  [Checksum Status: Good]
  Identifier (BE): 54564 (0xd524)
  Identifier (LE): 9429 (0x24d5)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Request frame: 4701]
  [Response time: 320.467 ms]
v ICMP Data: 69da76ef000498e508090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f3031323334353637
  Timestamp from icmp data: Apr 11, 2026 21:59:35.301285000 IST
  [Timestamp from icmp data (relative): 320.657000 milliseconds]
  > Data (48 bytes)

```

4. Examine the corresponding ping reply packet. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number and identifier fields?

The ICMP Type is **0** and the Code is **0**.

It contains the exact same structural fields as the request: **Checksum**, **Identifier**, **Sequence Number**, and the **Data payload**.

Just like the request packet, the Checksum, Identifier, and Sequence Number are each **2 bytes** long.

ICMP and Traceroute

```

[(base) adityagautam@Adityas-MacBook-Pro cs633a2 % traceroute www.inria.fr
traceroute to inria.fr (128.93.162.83), 64 hops max, 40 byte packets
 1 172.23.71.254 (172.23.71.254) 5.531 ms 8.417 ms 5.195 ms
 2 gateway.iitk.ac.in (172.31.1.251) 4.571 ms 4.838 ms 4.143 ms
 3 14.139.38.1 (14.139.38.1) 5.602 ms 4.522 ms 6.222 ms
 4 10.137.161.133 (10.137.161.133) 64.056 ms 64.132 ms 63.100 ms
 5 10.255.238.110 (10.255.238.110) 59.400 ms
   10.255.238.141 (10.255.238.141) 44.381 ms
   10.255.238.110 (10.255.238.110) 56.779 ms
 6 10.255.238.254 (10.255.238.254) 45.073 ms
   10.255.232.201 (10.255.232.201) 53.390 ms
   10.255.238.166 (10.255.238.166) 60.893 ms
 7 180.149.48.18 (180.149.48.18) 61.384 ms 51.304 ms 62.829 ms
 8 * * *
 9 180.149.48.29 (180.149.48.29) 191.607 ms 192.090 ms *
10 * 180.149.48.32 (180.149.48.32) 199.272 ms *
11 * lag-7-0.rt0.par.fr.geant.net (62.40.98.239) 268.033 ms *
12 lag-7-0.rt0.par.fr.geant.net (62.40.98.239) 209.740 ms
   renater-ap1-gw.rt0.par.fr.geant.net (62.40.124.70) 199.703 ms
   lag-7-0.rt0.par.fr.geant.net (62.40.98.239) 199.493 ms
13 hu0-3-0-41-ren-nr-orsay-rtr-091.noc.renater.fr (193.51.180.131) 200.895 ms 207.025 ms
   renater-ap1-gw.rt0.par.fr.geant.net (62.40.124.70) 211.376 ms
14 inria-rocquencourt-vl1631-te1-4-inria-rtr-021.noc.renater.fr (193.51.184.177) 286.695 ms 216.2
   60 ms
   hu0-3-0-41-ren-nr-orsay-rtr-091.noc.renater.fr (193.51.180.131) 218.149 ms
15 * inria-rocquencourt-vl1631-te1-4-inria-rtr-021.noc.renater.fr (193.51.184.177) 243.760 ms 229
   .449 ms
16 * * *
17 * * *
18 * * *

```

5. What is the IP address of your host? What is the IP address of the target destination host?

The source host's IP address is **172.23.68.32**. The target destination host (www.inria.fr) has the IP address **128.93.162.83**.

6. If ICMP sent UDP packets instead (as in Unix/Linux), would the IP protocol number still be 01 for the probe packets? If not, what would it be?

Because this capture was executed on macOS (a Unix-based system), the traceroute program is inherently sending **UDP packets**. The IP protocol number for these probe packets is not 01; it is **17** (which is **0x11** in hexadecimal) to denote UDP.

7. Examine the ICMP echo packet in your screenshot. Is this different from the ICMP ping query packets in the first half of this lab? If yes, how so?

Yes, it is fundamentally different. Because a Unix-based system was used, there are no "ICMP echo packets" sent as probes for the traceroute. Instead, the probes are **UDP packets** directed to **high destination port numbers** (e.g., Destination Port 33441), whereas the ping queries in the first half of the lab utilized native **ICMP headers**.

```
> Frame 1007: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0
> Ethernet II, Src: Cisco_e8:0d:42 (64:8f:3e:e8:0d:42), Dst: Apple_8b:47:44 (9c:3e:53:8b:47:44)
< Internet Protocol Version 4, Src: 14.139.38.1, Dst: 172.23.68.32
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 56
  Identification: 0x0000 (0)
  < 000. .... = Flags: 0x0
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 253
  Protocol: ICMP (1)
  Header Checksum: 0x9901 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 14.139.38.1
  Destination Address: 172.23.68.32
  [Stream index: 22]
< Internet Control Message Protocol
  > Type: Time-to-live exceeded (11)
  Code: 0 (Time to live exceeded in transit)
  Checksum: 0x000e [correct]
  [Checksum Status: Good]
  Unused: 00000000
  < Internet Protocol Version 4, Src: 172.23.68.32, Dst: 128.93.162.83
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 40
    Identification: 0xa51f (42271)
    > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    > Time to Live: 1
    Protocol: UDP (17)
    Header Checksum: 0x01be [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.23.68.32
    Destination Address: 128.93.162.83
    [Stream index: 19]
  < User Datagram Protocol, Src Port: 42264, Dst Port: 33441
    Source Port: 42264
    > Destination Port: 33441
    Length: 20
    Checksum: 0xc523 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 20]
```

8. Examine the ICMP error packet in your screenshot. It has more fields than the ICMP echo packet. What is included in those fields?

The ICMP error packet (Type 11, Time-to-live exceeded) includes the standard ICMP **Type**, **Code**, and **Checksum**. Additionally, its payload contains the **entire original IP header** of the probe packet that caused the error, along with the **first 8 bytes of the original datagram's payload** (which, in this specific trace, contains the **UDP header** with the source and destination port numbers).

9. Examine the last three ICMP packets received by the source host. How are these packets different from the ICMP error packets? Why are they different?

In this specific traceroute execution, the trace **timed out** (indicated by * * *) after hop 15 because the target network's firewall dropped the incoming UDP probe packets. Therefore, **no final ICMP packets were received**. In a completely open network, these final packets would differ from the error packets (Type 11) because they would be **Destination Unreachable / Port Unreachable (Type 3)**, signifying the packet reached the target host, but the specific UDP port was closed.

10. Within the tracert measurements, is there a link whose delay is significantly longer than others? Refer to the screenshot in Figure 4, is there a link whose delay is significantly longer than others? On the basis of the router names, can you guess the

location of the two routers on the end of this link?

Yes, there is a significant delay increase starting after hop 7 (180.149.48.18), where the delay jumps from around **60 ms** to over **190 ms** by hop 9, and over **200 ms** by hop 12. Based on the DNS name of the router at hop 12 (lag-7-0.rt0.par.fr.geant.net), the **.par.fr.** segment strongly suggests this router is located in **Paris, France**. The massive jump in delay represents the physical time it takes for the packet to cross the **transcontinental link** from the regional network in India to Europe.

Part 2: IP Lab

1. What is the IP address of your computer?

The IP address of the source computer is **172.23.68.32**.

2. Within the IP packet header, what is the value in the upper layer protocol field?

For the macOS traceroute probe packets sent out in this capture, the upper layer protocol field value is **17**, which corresponds to **UDP**.

3. How many bytes are in the IP header? How many bytes are in the payload of the IP datagram? Explain how you determined the number of payload bytes.

There are **20 bytes** in the IP header (indicated by the **Header Length** field).

For the standard UDP probe datagrams in the trace, the **Total Length** is 40 bytes. Therefore, the payload is **20 bytes**. This is determined by taking the Total Length of the IP datagram and subtracting the IP Header Length (**40 - 20 = 20**).

4. Has this IP datagram been fragmented? Explain how you determined whether or not the datagram has been fragmented.

No, the standard 40-byte probe datagram has not been fragmented. This is determined by observing that the **More fragments** flag is set to **0 (Not set)** and the **Fragment offset** is exactly **0**.

5. Which fields in the IP datagram always change from one datagram to the next within this series of ICMP messages sent by your computer?

The **Identification**, **Time to Live (TTL)**, and **Header Checksum** fields constantly change from one outgoing datagram to the next.

6. Which fields stay constant? Which of the fields must stay constant? Which fields must change? Why?

Stay Constant: **Version**, **Header Length**, **Source IP**, **Destination IP**, and **Protocol**. These *must* stay constant to ensure the datagram relies on the correct IPv4 format, maintains the correct end-to-end routing addresses, and correctly signals the upper-layer protocol (UDP).

Must Change: The **Identification** field (to uniquely identify each datagram for potential reassembly), the **TTL** field (which must systematically increment by 1 to discover successive traceroute hops), and the **Checksum** (which must be recalculated because the TTL and Identification values are changing).

7. Describe the pattern you see in the values in the Identification field of the IP datagram

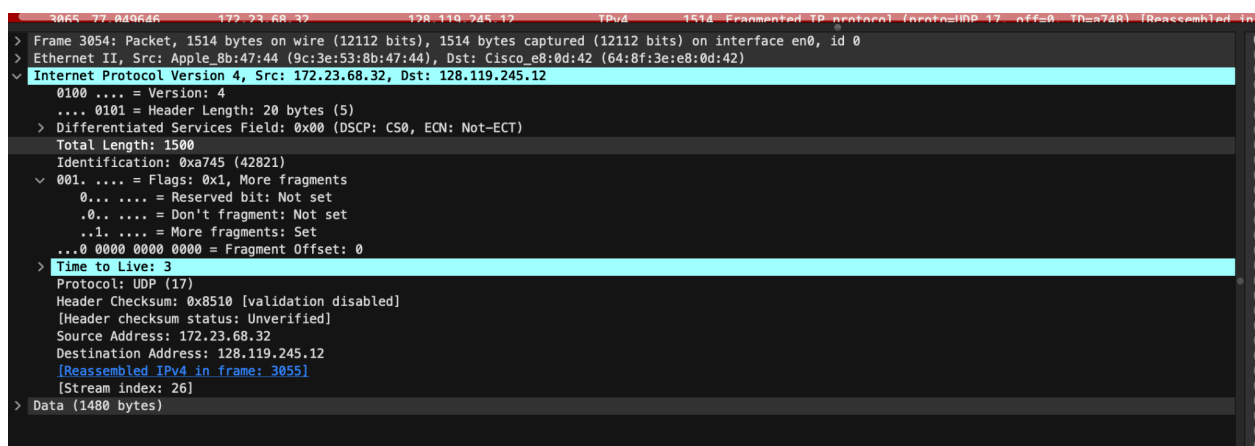
The values in the **Identification** field systematically **increment** for each newly generated outgoing datagram.

8. What is the value in the Identification field and the TTL field?

In the ICMP error replies returned by intermediate routers, the **Identification** field varies randomly depending on the sequential assignment state of the specific responding router. The **TTL** field generally arrives with a value derived from the router's **default starting TTL** (e.g., 255) decremented by the return hop count.

9. Do these values remain unchanged for all of the ICMP TTL-exceeded replies sent to your computer by the nearest (first hop) router? Why?

The **TTL** value remains unchanged because the topological distance (in network hops) from the first router back to the host is a constant **1 hop**. However, the **Identification** field changes because the router assigns a unique, newly generated ID for every discrete error packet it mints.



```
3065 77.8d96d6 172.23.68.32 128.119.245.12 IPv4 1514 Fragmented IP protocol (protocol: 17, offset: 10, 748) (Reassembled in
> Frame 3054: Packet, 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface en0, id 0
> Ethernet II, Src: Apple_8b:47:44 (9c:3e:53:8b:47:44), Dst: Cisco_e8:0d:42 (64:8f:3e:e8:0d:42)
> Internet Protocol Version 4, Src: 172.23.68.32, Dst: 128.119.245.12
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 1500
    Identification: 0xa745 (42821)
  > 001. .... = Flags: 0x1, More fragments
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..1. .... = More fragments: Set
    ...0 0000 0000 0000 = Fragment Offset: 0
  > Time to Live: 3
    Protocol: UDP (17)
    Header Checksum: 0x8510 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.23.68.32
    Destination Address: 128.119.245.12
    [Reassembled IPv4 in frame: 3055]
    [Stream index: 26]
  > Data (1480 bytes)
```

10. Find the first ICMP Echo Request message that was sent by your computer after you changed the Packet Size in pingplotter to be 2000. Has that message been fragmented

across more than one IP datagram?

(Note: Per macOS behavior, this is a UDP packet rather than an ICMP Echo Request). Yes, the message has been fragmented. Because the trace dictated a **2000-byte packet** and standard Ethernet restricts the **Maximum Transmission Unit (MTU)** to **1500 bytes**, the datagram had to be split.

11. Print out the first fragment of the fragmented IP datagram. What information in the IP header indicates that the datagram been fragmented? What information in the IP header indicates whether this is the first fragment versus a latter fragment? How long is this IP datagram?

Fragmentation is explicitly indicated by the **Flags** field, where the **More fragments** bit is marked as **Set (1)**.

It is identified as the first fragment because the **Fragment offset** is exactly **0**.

The **Total Length** of this specific IP datagram is **1500 bytes**.

```
> Frame 3055: Packet, 534 bytes on wire (4272 bits), 534 bytes captured (4272 bits) on interface en0, id 0
> Ethernet II, Src: Apple_8b:47:44 (9c:3e:53:8b:47:44), Dst: Cisco_e8:0d:42 (64:8f:3e:e8:0d:42)
< Internet Protocol Version 4, Src: 172.23.68.32, Dst: 128.119.245.12
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 520
    Identification: 0xa745 (42821)
  < 000. .... = Flags: 0x0
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..0. .... = More fragments: Not set
    ...0 0000 1011 1001 = Fragment Offset: 1480
  > Time to Live: 3
    Protocol: UDP (17)
    Header Checksum: 0xa82b [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.23.68.32
    Destination Address: 128.119.245.12
  > [2 IPv4 Fragments (1980 bytes): #3054(1480), #3055(500)]
    [Stream index: 26]
  < User Datagram Protocol, Src Port: 42812, Dst Port: 33443
    Source Port: 42812
    > Destination Port: 33443
    Length: 1980
    Checksum: 0x60da [unverified]
    [Checksum Status: Unverified]
    [Stream index: 71]
    [Stream Packet Number: 1]
  > [Timestamps]
  > UDP payload (1972 bytes)
  > Data (1972 bytes)
```

12. Print out the second fragment of the fragmented IP datagram. What information in the IP header indicates that this is not the first datagram fragment? Are the more fragments? How can you tell?

It is identified as a latter fragment because the **Fragment offset** is greater than 0 (specifically, **1480**, meaning 1480 bytes of payload precede this packet).

There are no more fragments for this datagram. This is confirmed by the **Flags** field, where the **More fragments** bit is marked as **Not set (0)**.

13. What fields change in the IP header between the first and second fragment?

Four fields change between the first and second fragment:

1. **Total Length:** Decreases from 1500 in the first fragment to 520 in the second fragment.
2. **Flags:** The "More fragments" bit shifts from **Set** to **Not set**.
3. **Fragment Offset:** Shifts from **0** to **1480**.
4. **Header Checksum:** Changes (e.g., from **0x8510** to **0xa82b**) because it is recalculated due to the modified header values.

14. How many fragments were created from the original datagram?

3 fragments are created for a 3500-byte datagram. With an Ethernet MTU of 1500 bytes, the maximum IP payload permitted per packet is **1480 bytes** (leaving 20 bytes for the standard IPv4 header). Sending roughly 3500 bytes of data requires splitting it across three pieces: **1480 bytes, 1480 bytes**, and the **remaining payload** in the final packet.

15. What fields change in the IP header among the fragments?

The same four fields change systematically across the entire fragment sequence: the **Total Length** (1500, 1500, and the remaining length respectively), the **Flags** (the first two will have "More fragments" **Set**; the final will be **Not set**), the **Fragment Offset** (0, 1480, and 2960), and the **Header Checksum**.

Question 3

PS: Consider a 2x2 switch with infinitely backlogged input queues. Destinations are randomly and uniformly chosen from $\{1, 2\}$ and are independent across packets. Let $d_i(t)$ for $(i=1,2)$ denote the destination of the head-of-line (HOL) packet of input queue i at time t .

- When $d_1(t) \neq d_2(t)$, both HOL packets are transferred.
- When $d_1(t) = d_2(t)$, one is randomly selected and transferred.

Part 1: Transition Matrix and Stationary Distribution

Let the state space of the Markov chain $d(t)$ be $S = \{11, 12, 21, 22\}$, where a state is represented by $(d_1(t), d_2(t))$. Let us analyze the transition probabilities from each state to the next state at time $t+1$.

1. Transitions from states 12 and 21 (No Conflict):

In these states, $d_1(t) \neq d_2(t)$. Both head-of-line packets are successfully transferred to their respective output ports. In the next time slot, two entirely new packets advance to the head of the queues. Because destinations are chosen randomly and uniformly from $\{1, 2\}$ and are independent, the next state can be any of the four states with equal probability ($1/2 \times 1/2 = 1/4$).

- $P(12 \rightarrow 11) = P(12 \rightarrow 12) = P(12 \rightarrow 21) = P(12 \rightarrow 22) = 1/4$
- $P(21 \rightarrow 11) = P(21 \rightarrow 12) = P(21 \rightarrow 21) = P(21 \rightarrow 22) = 1/4$

2. Transitions from state 11 (Conflict):

Both packets are destined for output port 1. One is randomly selected (each with probability $1/2$) and transferred.

- **Case A (Queue 1 is selected):** This happens with probability $1/2$. Queue 2 retains its blocked packet (destined for port 1). Queue 1 gets a new packet destined for port 1 (prob $1/2$) or port 2 (prob $1/2$). The resulting next state is 11 (prob $1/2 \times 1/2 = 1/4$) or 21 (prob $1/2 \times 1/2 = 1/4$).
- **Case B (Queue 2 is selected):** This happens with probability $1/2$. Queue 1 retains its blocked packet (destined for port 1). Queue 2 gets a new packet destined for port 1 (prob $1/2$) or port 2 (prob $1/2$). The resulting next state is 11 (prob $1/2 \times 1/2 = 1/4$) or 12 (prob $1/2 \times 1/2 = 1/4$).

Summing the probabilities for the next states from state 11:

- $P(11 \rightarrow 11) = 1/4 + 1/4 = 1/2$
- $P(11 \rightarrow 12) = 1/4$
- $P(11 \rightarrow 21) = 1/4$
- $P(11 \rightarrow 22) = 0$

3. Transitions from state 22 (Conflict):

By symmetry with state 11, the transition probabilities are:

- $P(22 \rightarrow 22) = 1/2$
- $P(22 \rightarrow 12) = 1/4$
- $P(22 \rightarrow 21) = 1/4$
- $P(22 \rightarrow 11) = 0$

Transition Matrix (P):

Ordering the states as (11, 12, 21, 22), we construct the transition matrix P:

$$P = \begin{bmatrix} 1/2 & 1/4 & 1/4 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 1/4 & 1/4 & 1/2 \end{bmatrix}$$

Stationary Distribution (π):

To find the stationary distribution $\pi = [\pi_{11}, \pi_{12}, \pi_{21}, \pi_{22}]$, we solve the global balance equation $\pi P = \pi$ subject to the normalization condition $\sum \pi_i = 1$.

Due to the symmetry of the transition matrix and the system, we can deduce that:

$$\pi_{11} = \pi_{22}$$

$$\pi_{12} = \pi_{21}$$

Let $\pi_{11} = a$ and $\pi_{12} = b$. Since $2a + 2b = 1$, we know $a + b = 1/2$.

Expanding the first column of the equation $\pi P = \pi$:

$$\pi_{11} = \pi_{11}(1/2) + \pi_{12}(1/4) + \pi_{21}(1/4) + \pi_{22}(0)$$

$$a = a(1/2) + b(1/4) + b(1/4)$$

$$a = a(1/2) + b(1/2)$$

$$a(1/2) = b(1/2) \implies a = b$$

Since $a = b$ and $a + b = 1/2$, it follows that $a = 1/4$ and $b = 1/4$.

Therefore, the stationary distribution is uniformly distributed across all states:

$$\pi = [1/4, 1/4, 1/4, 1/4]$$

Part 2: Average Throughput

The average throughput (T) of the switch is defined as the expected number of packets successfully transferred to the output ports per time slot in the steady state.

We compute this using the stationary distribution and the number of packets transferred in each specific state:

- In state **11** (Conflict): 1 packet is transferred.
- In state **12** (No Conflict): 2 packets are transferred.
- In state **21** (No Conflict): 2 packets are transferred.
- In state **22** (Conflict): 1 packet is transferred.

Taking the expectation:

$$T = \pi_{11}(1) + \pi_{12}(2) + \pi_{21}(2) + \pi_{22}(1)$$

$$T = (1/4)(1) + (1/4)(2) + (1/4)(2) + (1/4)(1)$$

$$T = \frac{1}{4} + \frac{2}{4} + \frac{2}{4} + \frac{1}{4}$$

$$T = \frac{6}{4} = 1.5$$

Conclusion:

The average throughput of this 2x2 switch under Head-of-Line (HOL) blocking is **1.5 packets**

per time slot (which equates to 0.75 packets per output port per time slot).